COMP3670 / COMP6670: Introduction to Machine Learning
2026 Semester 2 Assignment 3

Instructions

Please read all instructions before attempting the questions. Failure to follow the instructions may result in a mark penalty.

- **Deadline:** 28 October 2026, 11:59pm (Canberra time)
- **Submission:**
  - Submit your solutions to all questions in **a single PDF file** through Gradescope.
  - Start each of Question 1(a), Question 1(b), Question 2 and Question 3 on a new page.
  - You are encouraged to typeset your solutions using LaTeX. If you opt for handwriting, make sure that it is clear and easy to read. Unreadable writing may not be graded.
- **Grading:**
  - There are three questions, worth 15, 8 and 7 marks respectively, for a total of 30 marks.
  - After uploading your PDF file to Gradescope, **assign page numbers to each of the following four submission components:**
    - Question 1(a)
    - Question 1(b)
    - Question 2
    - Question 3

    A 10% penalty will be applied for each listed component for which you do not assign page numbers. A guide to assigning page numbers is available here.
  - Justify your answers and show all your steps clearly. Correct numerical answers without supporting justification or working will receive zero marks.
- **Academic Integrity:**
  - This is an **individual assignment**. You must work on your own.
  - The use of any generative AI tool is **not** allowed.
  - For more detailed guidelines on academic integrity, see Week 1 lecture slides and ANU webpage on academic integrity.

Question 1: Derivatives and gradient descent (15 marks)

Figure 1 shows the graph of the **derivative** f' of a twice differentiable function $f : \mathbb{R} \to \mathbb{R}$, and **not** the graph of f itself. The graph of f' is shown for $x \in [-2.5, 3.5]$ only, and outside this interval f' is positive and increases without bound as x moves away from it, both as $x \to -\infty$ and as $x \to +\infty$. Figure 2 zooms in on the same graph near $x = 1$ and near $x = 3$.

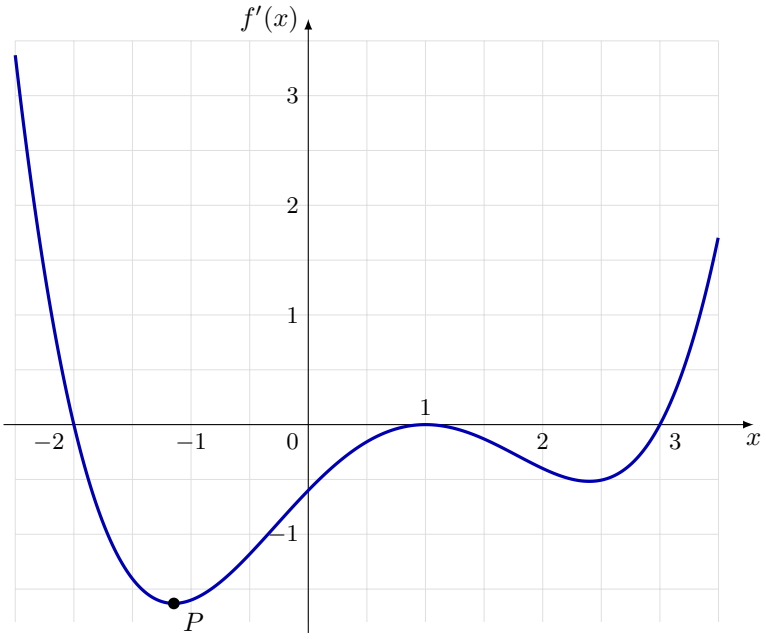
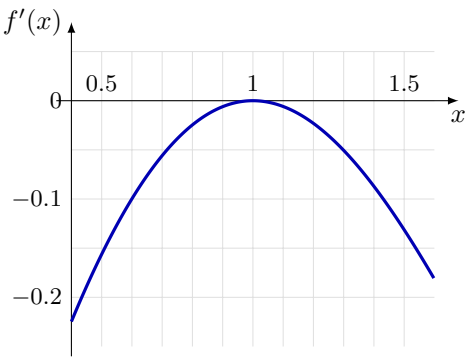
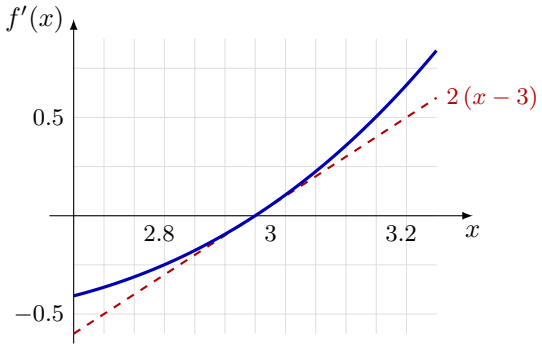


Figure 1: The graph of the derivative f'.



near $x = 1$



near $x = 3$, with the tangent line

Figure 2: The graph of f' zoomed in near $x = 1$ (left) and near $x = 3$ (right).

(a) Answer using only the graph of f', and justify every answer with what the graph shows.

(i) On which intervals is f increasing, and on which is it decreasing? **[2 marks]**

(ii) Find all the critical points of f. **[1 mark]**

(iii) Classify each critical point as a local minimum, a local maximum, or neither. **[3 marks]**

(iv) The point P is the lowest point of the graph of f'. Does f have a local minimum at this value of x? What changes in the shape of f at this value of x? **[1 mark]**

(b) To look for a minimum of f, we run gradient descent: from a starting value x_0 and with a learning rate $\alpha > 0$,

$$x_{t+1} \;=\; x_t - \alpha\, f'(x_t), \qquad t = 0, 1, 2, \ldots$$

(i) In which direction (left or right) does the first step move when $x_0 = 2.5$, when $x_0 = 3.4$ and when $x_0 = -2.5$, whatever the value of α? **[2 marks]**

(ii) Figure 2 shows that, near $x = 3$, the graph of f' is close to its tangent line $2\,(x-3)$. This line is exactly the derivative of the simpler function $g(x) = (x-3)^2$. In this item we run gradient descent on g instead of f, starting from $z_0 = 3.1$:

$$z_{t+1} \;=\; z_t - \alpha\, g'(z_t) \;=\; z_t - 2\alpha\,(z_t - 3), \qquad t = 0, 1, 2, \ldots$$

(A) Show that $z_{t+1} - 3 = (1 - 2\alpha)\,(z_t - 3)$, and hence that $z_t - 3 = 0.1\,(1-2\alpha)^t$.

(B) For $\alpha = 2$, $\alpha = 0.75$ and $\alpha = 0.25$, compute z_1, z_2 and z_3, and say whether z_t converges to 3 as $t \to \infty$.

(C) For which values of $\alpha > 0$ does z_t converge to 3?

(D) Gradient descent on f itself, started at $x_0 = 3.1$, gives the first iterates 3.044, 3.021, 3.010 when $\alpha = 0.25$; 2.931, 3.026, 2.986 when $\alpha = 0.75$; and 2.650, 3.536, -0.282 when $\alpha = 2$. Compare them with your answer to (B). Does g describe well how gradient descent behaves on f near $x = 3$?

[3 marks]

(iii) Starting from $x_0 = 0$ with $\alpha = 0.5$, the first iterates are 0.3, 0.452, 0.546, 0.610, 0.658, and the sequence converges. To which value does it converge? Is that value a local minimum of f? After gradient descent has converged, can we be sure that it has found a local minimum? Why? **[2 marks]**

(iv) What happens when $x_0 = -2.5$, for any $\alpha > 0$? What does this tell you about the global minimum of f? **[1 mark]**

Question 2: A hard-margin SVM with three datapoints (8 marks)

Let $d \geq 2$ be an integer. Consider the following labelled datapoints in $\mathbb{R}^d$:

$$\mathbf{x}_1 = \mathbf{0}, \quad y_1 = +1, \qquad \mathbf{x}_2 = \mathbf{u}, \quad y_2 = -1, \qquad \mathbf{x}_3 = \mathbf{v}, \quad y_3 = -1.$$

Assume that $\mathbf{u} \neq \mathbf{v}$. We seek the hard-margin SVM classifier with decision function $\mathbf{w}^\mathsf{T}\mathbf{x} + \beta$, where the positive side is assigned label $+1$.

The line segment joining $\mathbf{u}$ and $\mathbf{v}$ is

$$\{\mathbf{r}(t) = (1-t)\mathbf{u} + t\mathbf{v} : 0 \leq t \leq 1\}.$$

You may use the following geometric fact without proof: when this dataset is linearly separable, its maximum-margin separating hyperplane passes through $\mathbf{r}_*/2$ and is perpendicular to $\mathbf{r}_*$, where $\mathbf{r}_*$ is the point on the segment closest to the origin.

(a) Write down the hard-margin SVM minimisation problem in $\mathbf{w}$ and β, using the objective $\frac{1}{2}\|\mathbf{w}\|^2$ and explicitly giving the three constraints. **[1 mark]**

(b) Under what geometric condition does no hyperplane strictly separating the two classes exist? Express your answer in terms of the origin and the line segment joining $\mathbf{u}$ and $\mathbf{v}$, and briefly justify it. **[1 mark]**

(c) For parts (c) and (d), assume that the dataset is linearly separable. Give necessary and sufficient conditions for $\mathbf{u}$ to be the point on the segment closest to the origin, and for $\mathbf{v}$ to be that point. Express your conditions using $\|\mathbf{u}\|^2$, $\|\mathbf{v}\|^2$ and $\mathbf{u}^\mathsf{T}\mathbf{v}$, and justify your answers, including the equality cases. **[2 marks]**

(d) Now suppose that neither endpoint is closest to the origin, so that the closest point lies strictly between $\mathbf{u}$ and $\mathbf{v}$. Find $t_* \in (0,1)$ and the closest point $\mathbf{r}_* = (1-t_*)\mathbf{u} + t_*\mathbf{v}$. Hence find the optimal $\mathbf{w}_*$ and β_* in the scaling used in part (a), give the equation of the separating hyperplane, and find the geometric margin (the minimum distance from a training point to the hyperplane). You may express your answer using inner products of $\mathbf{u}$ and $\mathbf{v}$ and the vector $\mathbf{r}_*$. **[4 marks]**

Question 3: PCA and reconstruction (7 marks)

A dataset contains four datapoints in $\mathbb{R}^3$:

$$\mathbf{x}_1 = \begin{pmatrix} 6 \\ 4 \\ 4 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 4 \\ 6 \\ 0 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_4 = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}.$$

Apply PCA after centering the data, without rescaling any feature. Give exact answers and show your calculations.

(a) Find the sample mean $\bar{\mathbf{x}}$, the four centered datapoints, and the covariance matrix $\mathbf{C}$. **[2 marks]**

(b) You are given that the first principal direction is

$$\mathbf{p}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Verify that $\mathbf{p}_1$ is an eigenvector of $\mathbf{C}$, and find its corresponding eigenvalue. **[2 marks]**

(c) The second principal direction is

$$\mathbf{p}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Using only the first two principal directions, compute the two-dimensional PCA coordinates of each datapoint and reconstruct each datapoint in the original coordinates. **[3 marks]**